MILITARY SPECIFICATION

CHICKEN FILLET, CHUNKED AND FORMED, BREADED, PRECOOKED, FROZEN

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers frozen, boneless, round, breaded chicken fillets for use by the Department of Defense.

1.2 Classification. Chicken fillets shall be of the following types, as specified (see 6.1).

Type I - 3 ounce
Type II - 5 ounce

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-S-631	-	Spices, Ground and Whole, and Spice Blends
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1163	-	Boxes, Corrugated, Fiberboard, High Compression Strength, Weather Resistant, Wax Resin Impregnated

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8905

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Canned Foods for the Armed Forces

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Poultry Products Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Regulations Governing the Voluntary Grading of Poultry Products and Rabbit Products and U.S. Classes, Standards, and Grades

(Application for copies should be addressed to the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, South Building, 14th and Independence Avenue, S.W., Washington, DC 20250.)

Color Control on Breaded Chicken

(Application for copies should be addressed to the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, South Building, 14th and Independence Avenue, S.W., Washington, DC 20250.)

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2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Chicken. Only white (breast, first and second portion wing and scapula) chicken meat shall be used for this product. The chickens from which the white meat is derived shall be U.S. Grade A for fleshing and fat cover. The chickens shall be broilers/fryers and may be offered as carcasses or parts, bone-in or boneless. However, mechanically deboned (comminuted) meat shall be excluded from use in this product. The chicken shall be in the fresh-chilled state and shall be in excellent condition, i.e., lean and fat shall be of a color normally associated with fresh-chilled chickens and typical of chicken which has been properly handled; no odors foreign to fresh-chilled chicken shall be present. The chicken shall not have been frozen prior to use for this product.

3.2.2 Salt. Salt shall be non-iodized, white, refined sodium chloride with or without anti-caking agents.

3.2.3 Sodium tripolyphosphate. Sodium tripolyphosphate shall be Food Grade as permitted by the Poultry Products Inspection Regulations of the USDA.

3.2.4 White pepper. White pepper shall conform to type II of EE-S-631.

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3.2.5 Coating ingredients. Only commercially prepared coating ingredients (not artificially colored) commonly used for the type of product specified shall be permitted.

3.3 Preparation and processing (see 6.4).

3.3.1 Chicken preparation. The chicken shall be prepared as follows:

a. The chicken shall be skinned and raw boned. Mechanically deboned (comminuted) meat shall be excluded from use.

b. Only white (breast, first and second portion wing and scapula) meat shall be used. Other meat and carcass parts shall be excluded.

c. There shall be no bone, cartilage, ligaments, tendons, coarse connective tissue, blood clots, skin, or bruised or discolored meat exceeding 0.3 inch in any dimension.

d. The boned and trimmed chicken shall be reduced to chunk-like pieces by grinding once through a kidney shaped plate having openings measuring 2-1/2 inches or more in the least dimension, using a two-bladed knife.

3.3.2 Mixing. The chicken pieces shall be mechanically mixed with the following ingredients in the proportions indicated. Mixing shall continue for approximately 15 minutes or until the product becomes moderately tacky. The temperature of the chicken during mixing shall not exceed 45°F. The use of dry ice is permitted to control temperature during mixing.

Ingredients	Percent
Chicken white meat	99.16
Salt	0.50
Sodium tripolyphosphate	0.30
White pepper	0.04

3.3.3 Stuffing. The mixture shall be mechanically stuffed into suitable casings to form logs of such diameter that the finished product will conform to the diameter specified in 3.4 j. The logs shall be rapidly frozen to 0°F or below within 48 hours.

3.3.3.1 Log preparation time limitations. The grinding, mixing, stuffing, and placement of logs in the freezer shall be completed within one working day. The time period during which the logs may be held in the frozen state prior to tempering for slicing and processing into the final product shall not exceed 10 days.

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3.3.4 Tempering and slicing. All frozen logs shall be tempered to a temperature of 26° to 28°F and mechanically sliced into fillets of such thickness that the finished product will meet the requirements specified in 3.4. At no time during this operation shall the product show evidence of soft non-frozen areas more than 0.2 inches in depth on any surface. At no time shall the product be frozen and tempered more than once. The tempered product shall not be trimmed of softened (thawed) areas to comply with the 0.2 inch tolerance for this condition.

3.3.5 Coating. The fillets shall be uniformly coated with batter and breading. All batter and breading shall be applied by machine. Used breading may be sifted, when necessary, into clean containers for use the same day. Any unused batter or breading in the machine at the end of the day's production shall be discarded. Combined breading and batter on the fillet shall not exceed 23 percent by weight. The temperature of the batter used to coat the fillets shall not exceed 50°F during the coating operation.

3.3.6 Cooking. The coated fillets shall be fried in an approved vegetable oil containing an antioxidant which complies with the Poultry Products Inspection Regulations (9 CFR 381). The cooking shall be accomplished entirely by frying or by frying and then completing the cooking by use of microwaves, infrared energy, or hot air. The finished cooked chicken fillets shall have reached an internal temperature of 160°F and shall display a golden brown appearance of the coating.

3.3.7 Freezing. The cooked product shall be frozen as specified in either 3.3.7.1 or 3.3.7.2. The freezing process shall comply with the Poultry Products Inspection Regulations (9 CFR 381) except as otherwise specified herein. The use of dry ice is permitted to allow for rapid lowering of the fillet temperature.

3.3.7.1 Refrigeration and freezing method. Immediately after completion of cooking, the product shall be refrigerated and cooled to 40°F maximum. The product, at 40°F or lower, shall then be packed as specified in 5.1.1 and immediately subjected to freezing conditions so as to lower the temperature of the packed product to 0°F or lower within 72 hours. The time from completion of cooking until beginning of freezing shall not exceed 4 hours.

3.3.7.2 Individually quick frozen (IQF) method. Immediately after completion of cooking, the fillets shall be frozen by an IQF method that will rapidly change the moisture into ice crystals throughout the product and result in individual pieces that do not stick together after freezing. The frozen product shall be packed as specified in 5.1.1 and immediately placed under conditions of 0°F or lower.

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1. Weight of fillet shall be:

Type I - 3.0 to 3.5 ounces.
Type II - 5.0 to 5.5 ounces.

3.4.1 Palatability. The product shall be equal to or better than the approved preproduction sample in palatability and overall appearance.

3.5 Plant qualification. The poultry component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the USDA.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this document. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

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4.5.2.2 In-process examination for percent breading. Ten representative unbreaded fillets shall be weighed together immediately after the slicing operation. After cooking, ten cooked fillets from the area of production that the unbreaded fillets were drawn from shall be weighed together. The percent breading shall be calculated by the following formula and the results shall be reported to the nearest 1 percent.

$$\text{Percent breading} = \frac{\text{Weight of cooked fillets-weight of unbreaded fillets}}{\text{Weight of cooked fillets}} \times 100$$

Percent breading shall be calculated at least once per hour. If the result exceeds 23 percent breading, product from that hour's production shall be rejected.

4.5.3 Product inspection. The cooked product shall be examined for the defects listed in table II. The lot size shall be expressed in pounds. The sample unit shall be one fillet. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 2.5 for minor defects.

TABLE II. Product defects 1/ 2/ 3/

Category		Defect
Major	Minor	
101		Fillet separated into two or more pieces.
102		Presence of burnt (scorched) area greater than 0.5 inch in any dimension.
	201	Presence of more than one area devoid of batter/breading greater than 0.5 inch in any dimension.
	202	Presence of coating ridge (excess breading which projects at the edge of a fillet) larger than 0.5 inch.
	203	Presence of a lump (outcropping of coating on the surface of a fillet) greater than 0.3 inch in depth (thickness).
	204	Fillet not a uniformly brown color. 3/
	205	Weight of fillet less than 3.0 ounces or greater than 3.5 ounces for type I, or, weight of fillet less than 5.0 ounces or greater than 5.5 ounces for type II.
	206	Diameter of fillet less than 3.2 inches or more than 3.8 inches.

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4.5.5 Product frozen storage examination. Examination shall be made to determine conformance to the frozen storage requirements in 3.3.9. Any non-conformance revealed by actual examination or by review of records shall be cause for rejection of the involved product.

5. PACKAGING

5.1 Packing. Packing shall be level B.

5.1.1 Level B. Not more than 55 pounds of chicken fillets shall be packed in a snug-fitting fiberboard shipping container constructed and reinforced in accordance with style FTC, grade W5c or W5s of PPP-B-636. Alternatively, the shipping container shall be a wax-impregnated fiberboard box conforming to style RSC, class I, type SWCF1, grade 200 of PPP-B-1163. The shipping container shall be lined with a plastic bag of sufficient size to line the interior of the box completely and shall overlap not less than 5 inches along the center of the longer axis after the bag is folded down over the packed product. The product shall be arranged in layers with each layer no thicker than the thickness of one chicken fillet. Each layer of product shall be separated from each adjacent layer with a single sheet of polyethylene, not less than 1 inch longer than the length and width of the container. In addition, a sheet of polyethylene shall be placed between the inside bottom of the bag and the first layer and another placed on top of the top layer packed in the box. The polyethylene film for the bag and the separator material shall be of 0.0015 inch minimum thickness conforming to class 2 of L-P-378. Closure of the wax impregnated box shall be with hot-melt adhesive in accordance with the appendix of PPP-B-636. The wax-impregnated box shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636. Reinforcement of PPP-B-636 and PPP-B-1163 boxes shall be restricted to nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. All reinforcing bands shall be uncolored or of a color which contrasts with black.

5.2 Marking. Shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall be placed on the top of the container in capital letters between 1 and 1-1/2 inches high.

PERISHABLE - KEEP FROZEN
(0°F or BELOW)

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type of product required (see 1.2).
c. When a first article is required (see 3.1, 4.4, and 6.3).

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6.2 Award of contract. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.5.

6.3 First article. The item covered by this document requires first article test and approval under the appropriate provisions of DAR 7-104.55. The first article shall be a preproduction sample. The samples shall consist of 25 frozen, breaded, cooked chicken fillets which shall be submitted to the U.S. Army Natick Research and Development Center, ATTN: Food Engineering Laboratory, Office Code STRNC-WTP, Natick, MA 01760 for evaluation for acceptance. The contracting officer shall include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Other processes. Manufacturing processes and procedures which differ from those specified herein may be used provided the product manufactured therefrom is equivalent in composition and texture to that manufactured under the process procedures specified in this document. Alternative manufacturing processes must be approved by the U.S. Army Natick Research and Development Center prior to use. For this purpose twenty-five chicken fillets produced under the supervision of a USDA poultry grader (as evidenced by a proper USDA poultry grading certificate accompanying the fillets) should be sent to the following address together with completed details of the alternate process with which the chicken fillets were fabricated.

U.S. Army Natick Research and Development Center
Food Engineering Laboratory
Office Code STRNC-WTP
Natick, MA 01760

Custodian:

Army - GL

Review activities:

Army - MD, TS
DP - SS

Preparing activity:

Army - GL

Project No. 8905-A999

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44119(GL)	Chicken Fillet, Chunked and Formed, Breaded, Precooked, Frozen

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 APR 83

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